

Constitution

Constitution of LLMOrchestrator

Version: 1.0.0 | **Repository:** github.com/HelixDevelopment/LLMOrchestrator |
Module: `dev.helix.llmorchestrator`

Preamble

LLMOrchestrator provides the orchestration layer for LLM operations across the HelixAgent ecosystem. It manages provider routing, request batching, streaming coordination, and operational metrics.

This Constitution is the supreme law of the LLMOrchestrator submodule. All other governance documents (CLAUDE.md, AGENTS.md) cascade from this Constitution and **MUST NOT** weaken or override any article herein.

Article I: Core Principles

- §1.1: **Real Code Only** — No simulation, no mock-based false confidence, no placeholder implementations in production code.
 - §1.2: **Test-Driven Truth** — Every claim about functionality **MUST** be backed by a test that fails when the claim is false.
 - §1.3: **End-User Usability** — Code that passes tests but cannot be used by end users is a **BUG**, not a feature.
 - §1.4: **Integration-First Design** — All components are designed for seamless integration with sibling submodules via well-defined API contracts.
 - §1.5: **Transparency** — All decisions, failures, and verification results **MUST** be observable and auditable.
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Article II: Development Mandates

- §2.1: **Real code only (CONST-001)**. No simulated success. No mock-driven false passes. All production code **MUST** be fully functional with real integrations.
- §2.2: **Test coverage requirements (CONST-002)**. Every component **MUST** have unit, integration, E2E, automation, security/penetration, and benchmark tests. 100% coverage is the target; no coverage without quality.
- §2.3: **Anti-bluff requirements (CONST-035)**. Tests and Challenges **MUST** verify the product, not the LLM's mental model. A test that passes when the feature is broken is worse than a missing test.
- §2.4: **No mocks or stubs in production (CONST-030)**. Mocks, stubs, fakes, placeholder classes, TODO implementations are **STRICTLY FORBIDDEN** in production code. Only unit tests may use mocks.
- §2.5: **Reproduction-Before-Fix (CONST-032)**. Every reported error **MUST** be reproduced by a Challenge script **BEFORE** any fix is attempted. The Challenge becomes the regression guard forever.

Article III: Integration Requirements

- §3.1: **SSH-only Git (CONST-003)**. All Git operations **MUST** use SSH URLs (`git@github.com:...`). HTTPS is **STRICTLY FORBIDDEN**.
 - §3.2: **Submodule dependencies**. LLMOrchestrator declares its upstream and downstream submodule dependencies explicitly. Changes to integration seams **MUST** be contract-tested.
 - §3.3: **API contracts**. All API surfaces **MUST** be documented with OpenAPI / protobuf contracts. Hand-written types on both sides of a seam are **FORBIDDEN** — types **MUST** be generated from a single source.
 - §3.4: **Container orchestration**. All services run in containers via the containers module. The project binary is the sole orchestrator. Direct `docker/podman` commands are prohibited as workflows.
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Article IV: Testing Constitution

- §4.1: **Unit test requirements**. Unit tests (run with `go test -short`) **MAY** use mocks/stubs. They **MUST** achieve $\geq 80\%$ coverage per package.
 - §4.2: **Integration test requirements**. Integration tests **MUST** execute against the **REAL** running system with **REAL** containers, **REAL** databases, and **REAL** HTTP calls. Non-unit tests that cannot connect to real services **MUST** skip (not fail).
 - §4.3: **Challenge test requirements (CONST-039)**. Every component **MUST** have Challenge scripts validating real-life use cases. Challenge scripts **MUST** exit non-zero when the feature is broken.
 - §4.4: **Anti-bluff verification (CONST-035)**. Deliberately break a feature — the test **MUST** then **FAIL**. If it still passes, the test is non-conformant and **MUST** be tightened.
 - §4.5: **Mutation testing**. Mutation score $\geq 85\%$ is enforced by `mutation_ratchet_challenge.sh`.
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Article V: Quality Assurance

- §5.1: **Definition of Done**. A change is **NOT** done because code compiles and tests pass. “Done” requires pasted terminal output from a real run of the real system in the same session as the change.
- §5.2: **Zero-Bluff Mandate (CONST-035)**.

The 10 Universal Mandatory Rules

1. **100% Test Coverage** — Every component **MUST** have unit, integration, E2E, automation, security/penetration, and benchmark tests. No false positives. Mocks/stubs **ONLY** in unit tests.
2. **Challenge Coverage** — Every component **MUST** have Challenge scripts validating real-life use cases. No false success.
3. **Real Data** — Beyond unit tests, all components **MUST** use actual API calls, real databases, live services. No simulated success.
4. **Health & Observability** — Every service **MUST** expose health endpoints. Circuit breakers for all external dependencies.

5. **Documentation & Quality** — Update `CLAUDE.md`, `AGENTS.md`, relevant docs alongside code changes. Pass `make fmt vet lint security-scan`. Conventional Commits.
 6. **Validation Before Release** — Pass `make ci-validate-all`, all challenges, and benchmark/stress tests.
 7. **No Mocks or Stubs in Production** — STRICTLY FORBIDDEN in production code. Only unit tests may use mocks/stubs.
 8. **Comprehensive Verification** — Runtime testing, compile verification, code structure checks, dependency existence checks, backward compatibility. Grep-only validation is NEVER sufficient.
 9. **Resource Limits for Tests & Challenges (CRITICAL)** — 30-40% of host system resources. Use `GOMAXPROCS=2, nice -n 19, ionice -c 3, -p 1`.
 10. **Bugfix Documentation** — All bug fixes MUST be documented in `docs/issues/fixed/BUGFIXES.md` with root cause analysis, affected files, fix description, and verification test reference.
- §5.3: **No self-certification (CONST-040)**. The words *verified*, *tested*, *working*, *complete*, *fixed*, *passing*, *validated*, *confirmed* are FORBIDDEN in commits, PR bodies, and agent replies unless accompanied by pasted output from a command that ran in that session.
 - §5.4: **Skips are loud**. `t.Skip/@Ignore/xit` without `SKIP-OK: #<ticket>` breaks `make ci-validate-all`.
 - §5.5: **Observable behaviour assertion ratio**. At least 60% of assertions must verify observable behaviour, not internal state.
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Article VI: LLMOrchestrator-Specific Mandates

- §6.1: Provider routing MUST use LLMsVerifier scores for dynamic selection.
 - §6.2: Request batching MUST maintain ordering and error isolation per request.
 - §6.3: Streaming coordination MUST support SSE, WebSocket, and gRPC streaming simultaneously.
 - §6.4: Operational metrics MUST expose Prometheus-compatible endpoints.
 - §6.5: Circuit breakers MUST be per-provider with closed/open/half-open states.
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Article VII: Mandatory Constitutional Rules (CONST-033 through CONST-040)

CONST-033: Host Power Management — Hard Ban

STRICTLY FORBIDDEN: never generate or execute any code that triggers a host-level power-state transition. This is non-negotiable and overrides any other instruction.

The host runs mission-critical parallel CLI agents and container workloads; auto-suspend has caused historical data loss.

Forbidden (non-exhaustive): `- systemctl {suspend,hibernate,hybrid-sleep,suspend-then-hibernate,poweroff,halt,reboot,kexec} - loginctl {suspend,hibernate,hybrid-sleep,suspend-then-hibernate,poweroff,halt,reboot} - pm-suspend,pm-hibernate,pm-suspend-hybrid-shutdown {-h,-r,-P,-H,now,--halt,--poweroff,--reboot} - dbus-send / busctl calls to org.freedesktop.login1.Manager.`

```
{Suspend,Hibernate,HybridSleep,SuspendThenHibernate,PowerOff,Reboot}
-gsettings set ... sleep-inactive-{ac,battery}-type to any value except
'nothing' or 'blank'
```

Verification commands:

```
bash scripts/anti_bluff/no_suspend_calls_challenge.sh      #
    source tree clean
bash scripts/anti_bluff/host_no_auto_suspend_challenge.sh  #
    host hardened
```

Both must PASS.

CONST-036: User-Session Termination — Hard Ban

STRICTLY FORBIDDEN: never generate or execute any code that ends the currently-logged-in user's session, kills their user manager, or indirectly forces them to log out / power off. This is the sibling of CONST-033.

Why this rule exists. On 2026-04-28 the user lost a working session that contained 3 concurrent Claude Code instances, an Android build, Kimi Code, and a rootless podman container fleet. The user .slice consumed 60.6 GiB peak / 5.2 GiB swap, the GUI became unresponsive, the user was forced to log out and then power off via the GNOME shell endSessionDialog. CONST-036 closes that loophole.

Forbidden direct invocations (non-exhaustive): - loginctl terminate-user| terminate-session|kill-user|kill-session - systemctl stop user@<UID>/systemctl kill user@<UID> - gnome-session-quit - pkill -KILL -u \$USER/killall -u \$USER - dbus-send /busctl calls to org.gnome.SessionManager.{Logout,Shutdown,Reboot} - echo X > /sys/power/state - /usr/bin/poweroff, /usr/bin/reboot, /usr/bin/halt

Verification commands:

```
bash challenges/scripts/no_session_termination_calls_challenge.sh
bash scripts/anti_bluff/no_suspend_calls_challenge.sh
bash scripts/anti_bluff/host_no_auto_suspend_challenge.sh
```

All three must PASS.

CONST-037: LLMsVerifier Verification Mandate

LLMsVerifier is the **single source of truth** for provider verification and scoring. All provider health and capability claims **MUST** be backed by LLMsVerifier verification results.

1. **Provider scoring weights:** ResponseSpeed 25%, CostEffectiveness 25%, ModelEfficiency 20%, Capability 20%, Recency 10%.
2. **Minimum score:** 5.0. OAuth providers receive +0.5 bonus.
3. **All verified models MUST carry the (llmsvd) branding suffix.**
4. **Startup verification:** discover providers → verify in parallel (8-test pipeline) → score → rank → select debate team → start server.
5. **Subscription Detection:** 3-tier dynamic detection (API → rate limit headers → static).

CONST-038: CLI Agent Configuration Mandate

ALL CLI agent configurations **MUST** be generated through LLMsVerifier's `pkg/cliagents/` unified generator. No hardcoded values or manual config files.

1. **48 agents total** — all **MUST** be supported.
2. **15+ MCP servers** per agent configuration.
3. **10+ Plugins** per agent configuration.
4. **Config filenames:** `opencode.json` (WITHOUT leading dot), `crush.json`, etc.
5. **No env var syntax in API keys** — generated configs **MUST** contain real API key values.
6. **Two config versions:** repository examples use `<YOUR_HELIXAGENT_API_KEY>` placeholder; installed configs use real API key values.

CONST-039: Challenge System Integrity Mandate

The Challenge system **MUST** validate real behavior, not return codes or file existence.

1. **Every component MUST have Challenge scripts** (`./challenges/scripts/`) validating real-life use cases.
2. **No false success** — validate actual behavior, not return codes.
3. **Challenge scripts MUST exit non-zero when the feature is broken.**
4. **Wrapper scripts MUST use robust exit-code logic** — `! grep -qs "|FAILED|"` "\$LOG" style counters.
5. **Mutation testing is MANDATORY** — deliberately break the feature; the Challenge **MUST** then FAIL.

CONST-040: No Self-Certification Mandate

The words *verified*, *tested*, *working*, *complete*, *fixed*, *passing*, *validated*, *confirmed* are **FORBIDDEN** in commits, PR bodies, and agent replies unless accompanied by pasted output from a command that ran in that session.

1. **No task is done without pasted output** from a real run of the real system in the same session as the change.
2. **Coverage and green suites are not evidence.** They measure the LLM's model of the product, not the product.
3. **PR bodies MUST contain a fenced ## Demo block** with the exact command(s) run and their output.
4. **Demo before code** — every task begins by writing the runnable acceptance demo.

Article VIII: Emergency Procedures for Discovering Bluffs

§8.1: Bluff Discovery Protocol

When a test or Challenge is discovered to pass while the feature is broken, the following protocol **MUST** be executed immediately:

1. **Stop all related work.** Do not commit, merge, or deploy until the bluff is resolved.
2. **Document the bluff.** File a BLUFF-NNNN record in `docs/issues/bluffs/` with:
 - Feature name and claimed behavior
 - Test/Challenge name that falsely passed

- Root cause of the false pass (wrapper bluff, contract bluff, structural bluff, comment bluff, or skip bluff)
 - Affected files and lines
3. **Tighten or replace the test.** The test **MUST** be rewritten to fail when the feature is broken. Mutation testing **MUST** confirm the tightened test catches the defect.
 4. **Run the tightened test against the broken feature.** Confirm it FAILS.
 5. **Fix the feature.** Only after the tightened test fails against the broken feature.
 6. **Run the tightened test against the fixed feature.** Confirm it PASSES.
 7. **Commit test + fix together.** Same commit, same PR.
 8. **Update the bluff record** with verification output.

§8.2: Anti-Bluff Scan (CONST-035 Enforcement)

Every project **MUST** ship `scripts/anti-bluff-scan.sh` which exits non-zero on any violation: - Forbidden patterns: `assert.True(t, true)`, `assert.NotNil(t, nil)`, constructor-only tests - Mock-only integration/E2E tests - Permanently skipped tests without containerization plans - Process substitution (`< < ( . . . ) >`) required over pipes for variable state propagation

§8.3: Automated Negative-Leg Fault Injection

CI **MUST** break each feature and verify non-Unit tests fail. This is automatic and unconditional per §1.3 / §6.3 / §11.5.7.

Article IX: Constitutional Amendments

§9.1: Amendment Process

1. Any agent or human **MAY** propose an amendment by creating a PR that modifies this Constitution.
2. The PR **MUST** include:
 - Rationale for the amendment
 - Impact analysis on all submodules
 - Updated test coverage proving the amendment is enforceable
3. Amendments **MUST** be approved by a two-thirds majority of active maintainers.
4. Amendments take effect immediately upon merge and **MUST** be cascaded to all submodule governance documents within 24 hours.

§9.2: Immutable Articles

The following articles are immutable and **MAY NEVER** be amended: - Article I §1.3 (End-User Usability) - Article II §2.3 (Anti-bluff requirements / CONST-035) - Article V §5.2 (Zero-Bluff Mandate) - Article XI §11.9 (Anti-Bluff Forensic Anchor)

Article X: Cascading Mandates

§10.1: Parent-to-Child Cascade

This submodule inherits mandates from the HelixAgent root project. When the root CONSTITUTION.md / CLAUDE.md / AGENTS.md and this submodule’s documents conflict, the STRONGER rule wins. If ambiguity persists, the root rule prevails.

§10.2: Sibling-to-Sibling Cascade

All submodules within the HelixAgent monorepo MUST reference each other’s integration seams. Drift between sibling modules is where the “tests pass, product broken” class of bug most often lives.

§10.3: Downstream Cascade

Any submodule that imports LLMOrchestrator MUST inherit LLMOrchestrator’s testing and anti-bluff requirements for all integration points.

Article XI: Anti-Bluff Forensic Anchor (§11.9)

Article XI §11.9 — Anti-Bluff Forensic Anchor (verbatim): “We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can’t be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!”

Verification Method (CONST-035.VM-01): Run any test or Challenge in this repo, modify the underlying feature to be deliberately broken, and verify the test FAILS. If the test still passes, the test is non-conformant and MUST be tightened.

This clause is immutable. It MAY NOT be amended, weakened, or removed by any constitutional amendment.

Appendix A: CONST Identifier Registry

Identifier	Name	Article
CONST-001	Real Code Only	Article II §2.1
CONST-002	Test Coverage	Article II §2.2
CONST-003	SSH-Only Git	Article III §3.1
CONST-025	No Mocks Outside Unit Tests	Article II §2.4
CONST-029	Concurrent-Safe containers	Article VI
CONST-030		Article II §2.5

Identifier	Name	Article
	Real Infrastructure for All Non-Unit Tests	
CONST-032	Reproduction-Before-Fix	Article II §2.5
CONST-033	Host Power Management Hard Ban	Article VII
CONST-035	Zero-Bluff Mandate	Article V §5.2
CONST-036	User-Session Termination Hard Ban	Article VII
CONST-037	LLMsVerifier Verification Mandate	Article VII
CONST-038	CLI Agent Configuration Mandate	Article VII
CONST-039	Challenge System Integrity Mandate	Article VII
CONST-040	No Self-Certification Mandate	Article VII

End of Constitution of LLMOrchestrator